

Course Information

Course Number: CSCE 636
Course Title: Deep Learning
Section: 700
Time: Tuesdays and Thursdays (online)
Location: online
Credit Hours: 3

Instructor Details

Instructor: Anxiao Jiang
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Office Hours: 11:00-11:30am on Mondays via zoom

Course Description

The course covers fundamental techniques and best practices in deep learning, including the mathematical building blocks of neural networks, programming languages for deep learning, deep learning for classification and regression, fundamental machine learning techniques for deep learning, universal workflow of deep learning, deep learning for computer vision, deep learning for time-series and text, generative deep learning, and hyper-parameter optimization.

Course Prerequisites

Graduate Standing, python language, machine learning

Special Course Designation

N/A

Course Learning Outcomes

Upon finishing this class, students should be able to:

- 1. Develop deep learning models for various classification, regression and generation tasks*
- 2. Design algorithms and software for training deep learning models*
- 3. Analyze the performance of deep learning models*
- 4. Design hyper-parameter tuning and regularization techniques for optimizing models*

Textbook and/or Resource Materials

Recommended Textbooks

- Deep Learning with Python, Second Edition, By Francois Chollet, Manning Publications, December 2021
- Deep Learning: Foundations and Concepts, By Christopher M. Bishop and Hugh Bishop, Springer, November 2023.

Grading Policy

Six homework assignments: 25%

Three projects: 25% each (75% total)

Grading Scale: A = [90, 100], B = [80, 90), C = [70, 80), D = [60, 70), F = [0, 60)

Late Work Policy

With the exception of excused absences (see Student Rule 7), no homework or project submitted after the established deadline will be accepted.

Course Schedule

Week	Lecture Topics	Reading list	Notes and Assignments
1	Mathematical Building Blocks of Deep Learning	Chapter 1-2	
2	Programming Languages for Deep Learning	Chapter 3	
3	Deep Learning for Classification and Regression	Chapter 4	Project 1
4	Fundamental Machine Learning Techniques for Deep Learning	Chapter 5	
5	Universal Workflow of Deep Learning	Chapter 6	Project 2
6	Dive Deep into Deep Learning Programming	Chapter 7	
7	Convolutional Neural Networks	Chapter 8	Project 3
8	Fundamental Deep Learning for Computer Vision	Chapter 8	
9	Advanced Deep Learning for Computer Vision	Chapter 9	
10	Deep Learning for Time Series; LSTM and GRU Models	Chapter 10	
11	Deep Learning for Text; Self-Attention Mechanism and Transformer Models	Chapter 11	
12	Generative Deep Learning	Chapter 12	
13	Auto-Encoder and Generative-Adversarial Network Models	Chapter 12	
14	Hyper-Parameter Optimization for Deep Learning	Chapter 13	

University Policies

Attendance Policy

The university views class attendance and participation as an individual student responsibility. Students are expected to attend class and to complete all assignments.

Please refer to [Student Rule 7](#) in its entirety for information about excused absences, including definitions, and related documentation and timelines.

Makeup Work Policy

Students will be excused from attending class on the day of a graded activity or when attendance contributes to a student's grade, for the reasons stated in Student Rule 7, or other reason deemed appropriate by the instructor.

Please refer to [Student Rule 7](#) in its entirety for information about makeup work, including definitions, and related documentation and timelines.

Absences related to Title IX of the Education Amendments of 1972 may necessitate a period of more than 30 days for make-up work, and the timeframe for make-up work should be agreed upon by the student and instructor" ([Student Rule 7, Section 7.4.1](#)).

"The instructor is under no obligation to provide an opportunity for the student to make up work missed because of an unexcused absence" ([Student Rule 7, Section 7.4.2](#)).

Students who request an excused absence are expected to uphold the Aggie Honor Code and Student Conduct Code. (See [Student Rule 24](#).)

Academic Integrity Statement and Policy

"An Aggie does not lie, cheat or steal, or tolerate those who do."

"Texas A&M University students are responsible for authenticating all work submitted to an instructor. If asked, students must be able to produce proof that the item submitted is indeed the work of that student. Students must keep appropriate records at all times. The inability to authenticate one's work, should the instructor request it, may be sufficient grounds to initiate an academic misconduct case" ([Section 20.1.2.3, Student Rule 20](#)).

You can learn more about the Aggie Honor System Office Rules and Procedures, academic integrity, and your rights and responsibilities at aggiehonor.tamu.edu.

Americans with Disabilities Act (ADA) Policy

Texas A&M University is committed to providing equitable access to learning opportunities for all students. If you experience barriers to your education due to a disability or think you may have a disability, please contact Disability Resources in the Student Services Building or at (979) 845-1637 or visit disability.tamu.edu. Disabilities may include, but are not limited to attentional, learning, mental health, sensory, physical, or chronic health conditions. All students are encouraged to discuss their disability related needs with Disability Resources and their instructors as soon as possible.

Title IX and Statement on Limits to Confidentiality

Texas A&M University is committed to fostering a learning environment that is safe and productive for all. University policies and federal and state laws prohibit gender-based discrimination and sexual harassment, including sexual assault, sexual exploitation, domestic violence, dating violence, and stalking.

With the exception of some medical and mental health providers, all university employees (including full and part-time faculty, staff, paid graduate assistants, student workers, etc.) are Mandatory Reporters and must report to the Title IX Office if the employee experiences, observes, or becomes aware of an incident that meets the following conditions (see [University Rule 08.01.01.M1](#)):

- The incident is reasonably believed to be discrimination or harassment.
- The incident is alleged to have been committed by or against a person who, at the time of the incident, was (1) a student enrolled at the University or (2) an employee of the University.

Mandatory Reporters must file a report regardless of how the information comes to their attention – including but not limited to face-to-face conversations, a written class assignment or paper, class discussion, email, text, or social media post. Although Mandatory Reporters must file a report, in most instances, a person who is subjected to the alleged conduct will be able to control how the report is handled, including whether or not to pursue a formal investigation. The University's goal is to make sure you are aware of the range of options available to you and to ensure access to the resources you need.

Students wishing to discuss concerns in a confidential setting are encouraged to make an appointment with [Counseling and Psychological Services](#) (CAPS).

Students can learn more about filing a report, accessing supportive resources, and navigating the Title IX investigation and resolution process on the University's [Title IX webpage](#).

Statement on Mental Health and Wellness

Texas A&M University recognizes that mental health and wellness are critical factors that influence a student's academic success and overall wellbeing. Students are encouraged to engage in healthy self-care by utilizing the resources and services available from Counseling & Psychological Services (CAPS). 24-hour emergency help is available through the National Suicide Prevention Hotline (800-273-8255) or at suicidepreventionlifeline.org.